A Projection Lemma and a Nearly Unimodular Special Case for Determinant-Only Flatness

Draft note

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Abstract

This note concerns the open question, raised in the context of determinant-parameterized flatness bounds, whether the lattice width of a full-dimensional lattice-free rational polyhedron

$$P(A,b) = \{x \in \mathbb{R}^n : Ax \le b\}, \qquad A \in \mathbb{Z}^{m\times n}, \quad b \in \mathbb{Z}^m,$$

can be bounded by a function of the largest full-rank subdeterminant

$$\Delta_n(A) = \max\{|\det B| : B \text{ is an } n \times n \text{ submatrix of } A\}$$

alone, independent of the dimension. Throughout this note, "lattice-free" means $P(A,b)\cap\mathbb{Z}^n = \varnothing$, matching the stated open problem.

The note does *not* solve the open problem. It gives partial progress in three directions. First, it proves a projection lemma: if $z \in \mathbb{Z}^n$ is primitive and $Az \in \{-1,0,1\}^m$, then projection along z preserves lattice-freeness in the quotient lattice and cannot increase the relevant upper bound on lattice width. Second, it proves a dimension-free width bound for matrices that become totally unimodular after deleting a bounded number of variables. Third, it gives an explicit example showing that the naive Fourier–Motzkin projection along such a direction can increase the full-minor determinant parameter. Thus the projection lemma identifies a plausible route, but the missing determinant-control step remains essential.

1 The determinant-only flatness question

For a convex set $K \subseteq \mathbb{R}^n$ and a nonzero integral covector $c \in \mathbb{Z}^n \setminus \{0\}$, define

$$w_c(K) := \sup_{x\in K} c^\top x - \inf_{x\in K} c^\top x, \qquad \operatorname{width}(K) := \inf_{c\in\mathbb{Z}^n\setminus\{0\}} w_c(K).$$

If $A \in \mathbb{Z}^{m\times n}$ has full column rank, set

$$\Delta_n(A) := \max\{|\det B| : B \text{ is an } n \times n \text{ submatrix of } A\}.$$

Problem 1.1 (determinant-only flatness)**.** Does there exist a function $f : \mathbb{Z}_{\ge 1} \to \mathbb{R}_{\ge 0}$ such that, for every n, every full-column-rank $A \in \mathbb{Z}^{m\times n}$, and every $b \in \mathbb{Z}^m$, if

$$P(A,b) := \{x \in \mathbb{R}^n : Ax \le b\}$$

is full-dimensional and satisfies $P(A,b) \cap \mathbb{Z}^n = \varnothing$, then

$$\operatorname{width}(P(A,b)) \le f(\Delta_n(A))?$$

The classical flatness theorem gives a bound depending on n. Celaya, Kuhlmann, Paat, and Weismantel proved a facet-width bound linear in n and $\Delta_n(A)$, and asked whether the dimension dependence can be removed. Henk, Kuhlmann, and Weismantel studied closely related determinant-only width questions and proved dimension-free results for some special classes, while connecting the general problem to Hilbert-basis height phenomena.

The results below should be read as partial progress toward Problem 1.1, not as a resolution.

2 Preliminaries on quotient lattices

Let $z \in \mathbb{Z}^n$ be primitive. Then z can be completed to a unimodular basis of $\mathbb{Z}^n$. Equivalently, there exists $U \in \mathrm{GL}_n(\mathbb{Z})$ whose last column is z. The quotient map

$$\pi : \mathbb{R}^n \longrightarrow \mathbb{R}^n/\mathbb{R}z$$

sends $\mathbb{Z}^n$ onto the quotient lattice

$$\Lambda_z := \pi(\mathbb{Z}^n).$$

After the unimodular change of variables $x = U(y,t)$, with $y \in \mathbb{R}^{n-1}$ and $t \in \mathbb{R}$, the quotient lattice is identified with $\mathbb{Z}^{n-1}$.

3 Projection along a unit-slack direction

The following lemma is the main elementary observation.

Lemma 3.1 (unit-slack projection preserves lattice-freeness)**.** *Let*

$$P = \{x \in \mathbb{R}^n : Ax \le b\}, \qquad A \in \mathbb{Z}^{m\times n}, \quad b \in \mathbb{Z}^m.$$

Let $z \in \mathbb{Z}^n$ be primitive and suppose that

$$Az \in \{-1,0,1\}^m.$$

Let $\pi : \mathbb{R}^n \to \mathbb{R}^n/\mathbb{R}z$ be the quotient map and let $\Lambda_z = \pi(\mathbb{Z}^n)$. If $P \cap \mathbb{Z}^n = \varnothing$, then

$$\pi(P) \cap \Lambda_z = \varnothing.$$

Proof. Choose $U \in \mathrm{GL}_n(\mathbb{Z})$ whose last column is z, and write $x = U(y,t)$ with $y \in \mathbb{R}^{n-1}$ and $t \in \mathbb{R}$. Since unimodular changes preserve integer points, it is enough to work in these coordinates. The transformed inequalities have the form

$$p_i^\top y + q_i t \le b_i, \qquad p_i \in \mathbb{Z}^{n-1}, \quad q_i \in \{-1,0,1\}.$$

Indeed, the last coefficient in the ith transformed row is $a_i^\top z = (Az)_i$.

Suppose, toward a contradiction, that an integer point $y \in \mathbb{Z}^{n-1}$ belongs to the projection. Then there exists some real t satisfying all inequalities, so the fiber over y is a nonempty interval cut out by

$$t \le b_i - p_i^\top y \quad (q_i = 1), \qquad t \ge p_i^\top y - b_i \quad (q_i = -1),$$

together with the constraints with $q_i = 0$, which do not involve t. Since b_i and $p_i^\top y$ are integers, all finite upper and lower endpoints of this interval are integers. A nonempty interval in $\mathbb{R}$ whose finite endpoints are integral contains an integer point. Hence there exists $t \in \mathbb{Z}$ such that (y,t) satisfies all transformed inequalities. Transforming back gives an integer point of P, a contradiction. $\square$

Lemma 3.2 (width monotonicity under quotient projection). *In the setting of Lemma 3.1,*

$$\mathrm{width}(P) \leq \mathrm{width}(\pi(P)),$$

where the width of $\pi(P)$ is computed with respect to the quotient lattice Λ_z.

Proof. Every nonzero integral covector c on the quotient lattice Λ_z pulls back to a nonzero integral covector $c \circ \pi$ on $\mathbb{Z}^n$. Moreover,

$$w_{c\circ\pi}(P) = w_c(\pi(P)).$$

Taking the infimum over all nonzero integral covectors on the quotient gives the asserted inequality. □

4 What a projection proof would still need

A theorem of Kuhlmann and Weismantel gives a determinant-only threshold for the shortest nonzero vector of the lattice $A\mathbb{Z}^n$ in the infinity norm. We use the following consequence.

Theorem 4.1 (Kuhlmann–Weismantel threshold, quoted form). *For every $D \geq 1$ there is a number $\theta(D)$ such that the following holds. If $A \in \mathbb{Z}^{m\times r}$ has full column rank, $\Delta_r(A) \leq D$, and $r > \theta(D)$, then there exists a nonzero $z \in \mathbb{Z}^r$ with*

$$\|Az\|_\infty = 1.$$

After dividing z by the greatest common divisor of its coordinates, one may take z primitive, and then $Az \in \{-1, 0, 1\}^m$.

Combining Theorem 4.1 with Lemmas 3.1 and 3.2 suggests a natural strategy: repeatedly project along primitive directions z satisfying $Az \in \{-1, 0, 1\}^m$ until the dimension is bounded solely in terms of $\Delta_n(A)$, and then apply the ordinary flatness theorem in the remaining bounded dimension.

The following proposition formalizes exactly what additional determinant control would be sufficient. It is intentionally conditional; the hypothesis is the missing hard part.

Proposition 4.2 (conditional projection route). *Assume the following uniform determinant-stability hypothesis. For every initial determinant bound Δ there exists a number $D = D(\Delta)$ such that, starting from any integral system $Ax \leq b$ with $\Delta_n(A) \leq \Delta$, every polyhedron obtained by any finite sequence of unit-slack quotient projections as in Lemma 3.1 admits an integral inequality description whose full-rank minors are bounded by D.*

Then Problem 1.1 has a positive answer, with for instance

$$\mathrm{width}(P(A, b)) \leq \mathrm{Flt}(\theta(D(\Delta_n(A))))$$

whenever $P(A, b)$ is lattice-free.

Proof. Let $P_0 = P(A, b)$ and let $D = D(\Delta_n(A))$ be the determinant bound supplied by the hypothesis. Suppose the current projected polyhedron P_j has dimension $r_j > \theta(D)$ and has an integral description $A_j x \leq b_j$ with $\Delta_{r_j}(A_j) \leq D$. By Theorem 4.1, there is a primitive $z_j \in \mathbb{Z}^{r_j}$ with $A_j z_j \in \{-1, 0, 1\}^{m_j}$. Lemma 3.1 says that projection along z_j preserves lattice-freeness in the quotient lattice, and Lemma 3.2 says that pulling back a quotient width direction gives an upper bound for the previous width. By the determinant-stability hypothesis, the new projected polyhedron again has an integral description with full-rank minors bounded by D.

Iterating, the dimension strictly decreases at each step, and the process stops once the dimension is at most $\theta(D)$. The classical flatness theorem in dimension at most $\theta(D)$ gives width at most $\mathrm{Flt}(\theta(D))$ for the final projected lattice-free polyhedron. Pulling the width direction back through the sequence of quotient maps gives the same upper bound for the original $P(A,b)$. $\square$

Remark 4.3. A one-step estimate of the form "the next projected description has determinant parameter at most $\Phi(\Delta)$" is not by itself enough, because repeated projection could lead to iterates $\Phi^j(\Delta)$ depending on the original dimension. Proposition 4.2 requires determinant control in terms of the original Δ throughout the whole sequence.

5 A nearly totally unimodular special case

The next theorem is an unconditional partial result. It gives a determinant-free reduction whenever all but q variables occur through a totally unimodular matrix. Thus it gives a determinant-only width bound for any class in which q is bounded by a function of $\Delta_n(A)$.

Theorem 5.1 (bounded non-TU variable dimension)**.** *Let $A \in \mathbb{Z}^{m\times n}$. Suppose that, after a unimodular change of variables, one can write*

$$A = [T\ C], \qquad T \in \mathbb{Z}^{m\times(n-q)}, \quad C \in \mathbb{Z}^{m\times q},$$

where T is totally unimodular. Let

$$P = \{(u,v) \in \mathbb{R}^{n-q} \times \mathbb{R}^q : Tu + Cv \le b\}, \qquad b \in \mathbb{Z}^m,$$

be full-dimensional and lattice-free. Then

$$\mathrm{width}(P) \le \mathrm{Flt}(q).$$

Consequently, if this representation exists with $q \le h(\Delta_n(A))$ for some function h, then this class satisfies Problem 1.1 with

$$\mathrm{width}(P) \le \mathrm{Flt}(h(\Delta_n(A))).$$

Proof. Let

$$Q := \{v \in \mathbb{R}^q : \exists u \in \mathbb{R}^{n-q} \text{ such that } Tu + Cv \le b\}$$

be the projection of P onto the v-coordinates. We first show that $Q \cap \mathbb{Z}^q = \varnothing$.

Suppose, toward a contradiction, that $v \in Q \cap \mathbb{Z}^q$. Then the fiber

$$F_v := \{u \in \mathbb{R}^{n-q} : Tu \le b - Cv\}$$

is nonempty, and its right-hand side $d := b - Cv$ is integral. Choose some $u_0 \in F_v$ and choose an integer $M > \|u_0\|_\infty$. Then

$$F_v \cap [-M, M]^{n-q}$$

is a nonempty bounded polyhedron described by

$$Tu \le d, \qquad u \le M\mathbf{1}, \qquad -u \le M\mathbf{1}.$$

The matrix obtained by appending rows of I and $-I$ to a totally unimodular matrix is again totally unimodular. Therefore this bounded polyhedron has an integral vertex. That vertex lies in F_v, so $F_v \cap \mathbb{Z}^{n-q} \neq \varnothing$. Hence P contains an integer point, contradicting lattice-freeness.

Thus Q is lattice-free in $\mathbb{R}^q$. By the classical flatness theorem, $\operatorname{width}(Q) \le \operatorname{Flt}(q)$. If $c \in \mathbb{Z}^q \setminus \{0\}$ is a width direction for Q, then $(0, c) \in \mathbb{Z}^n \setminus \{0\}$ satisfies

$$w_{(0,c)}(P) = w_c(Q).$$

Hence $\operatorname{width}(P) \le \operatorname{width}(Q) \le \operatorname{Flt}(q)$. $\square$

Remark 5.2. For $q = 0$, the same fiber argument says that no nonempty lattice-free polyhedron of the displayed form exists: a nonempty system $Tu \le b$ with T totally unimodular and b integral contains an integer point.

6 The naive projection does not preserve the determinant parameter

The projection lemma is useful only if one can control the determinant parameter after projection. The next example shows that the most direct Fourier–Motzkin description need not preserve the same determinant bound.

Proposition 6.1 (Fourier–Motzkin projection can increase Δ). *Let*

$$A = \begin{pmatrix} 0 & -2 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & -1 \\ 1 & 0 & -1 \\ -1 & 0 & 1 \\ -3 & -3 & -1 \end{pmatrix}.$$

Then $\Delta_3(A) = 11$. For $z = e_3$ one has

$$Az = (1, -1, -1, -1, 1, -1)^\top \in \{-1, 1\}^6.$$

The standard Fourier–Motzkin projection eliminating the third coordinate has a valid row matrix

$$A' = \begin{pmatrix} 0 & 0 \\ -1 & 2 \\ 0 & -3 \\ -1 & -1 \\ 1 & -2 \\ 0 & 0 \\ -3 & -5 \\ -4 & -3 \end{pmatrix},$$

and $\Delta_2(A') = 12 > 11$.

Proof. A direct computation of the $\binom{6}{3} = 20$ full-rank minors of A gives

I	123	124	125	126	134	135	136	145	146	156
$\det A_I$	0	0	0	0	3	-3	-9	0	-11	11

and

I	234	235	236	245	246	256	345	346	356	456
$\det A_I$	-3	3	9	0	11	-11	0	-1	1	0.

Thus the maximum absolute value is 11.

Now write each row of A as (p_i, q_i) with $p_i \in \mathbb{Z}^2$ and $q_i \in \{\pm 1\}$. Rows with $q_i = 1$ give upper bounds on the eliminated variable, and rows with $q_j = -1$ give lower bounds. Combining a lower bound with an upper bound gives the projected row $p_i + p_j$. The two upper rows are

$$(0, -2), \qquad (-1, 0),$$

and the four lower rows are

$$(0, 2), \qquad (0, -1), \qquad (1, 0), \qquad (-3, -3).$$

The eight pairwise sums are precisely the displayed rows of A', up to ordering.

The two zero rows do not affect $\Delta_2(A')$. For the six nonzero row types

$$\begin{aligned} r_1 &= (-1, 2), & r_2 &= (0, -3), & r_3 &= (-1, -1), \\ r_4 &= (1, -2), & r_5 &= (-3, -5), & r_6 &= (-4, -3), \end{aligned}$$

the absolute pair determinants are

$\lvert\det(r_i, r_j)\rvert$	r_1	r_2	r_3	r_4	r_5	r_6
r_1	0	3	3	0	11	11
r_2		0	3	3	9	12
r_3			0	3	2	1
r_4				0	11	11
r_5					0	11
r_6						0.

Hence $\Delta_2(A') = 12$. $\square$

Remark 6.2. Proposition 6.1 does not disprove Problem 1.1. It also does not rule out a more sophisticated projected description with determinant parameter bounded by a function of the original determinant. It only shows that the raw Fourier–Motzkin projection cannot be used as a determinant-preserving black box.

7 Two simple non-counterexamples

The following examples explain why some tempting high-width constructions do not refute Problem 1.1.

Example 7.1 (a half-integral simplex)**.** For $n \geq 2$, define

$$P_n = \left\{ x \in \mathbb{R}^n : x_i \geq \frac{1}{2} \ (i = 1, \ldots, n), \quad \sum_{i=1}^{n} x_i \leq n - \frac{1}{2} \right\}.$$

Then $P_n \cap \mathbb{Z}^n = \varnothing$: if $x \in \mathbb{Z}^n$ and $x_i \geq 1/2$ for all i, then $x_i \geq 1$ for all i, so $\sum_i x_i \geq n$, contradicting the last inequality.

The vertices of P_n are

$$\frac{1}{2}\mathbf{1} \quad \text{and} \quad \frac{1}{2}\mathbf{1} + \frac{n-1}{2} e_j \qquad (j = 1, \ldots, n).$$

Thus P_n is a translate of $\frac{n-1}{2}\,\mathrm{conv}(0, e_1, \dots, e_n)$, and since the standard simplex has lattice width 1,

$$\mathrm{width}(P_n) = \frac{n-1}{2}.$$

However, an integral description is

$$-2x_i \le -1 \quad (i = 1, \dots, n), \qquad 2\sum_{i=1}^{n} x_i \le 2n - 1.$$

Every nonzero full-rank determinant of this inequality matrix has absolute value 2^n, so

$$\Delta_n(A_n) = 2^n.$$

The example has growing width, but its determinant parameter grows exponentially with the dimension.

Example 7.2 (interior-free is not lattice-free in this problem)**.** The simplex

$$S_n = \{x \in \mathbb{R}^n : x_i \ge 0 \ (i = 1, \dots, n), \quad \sum_i x_i \le n\}$$

has no integer point in its interior after suitable variants and has large natural width, but it contains many boundary lattice points. Since Problem 1.1 uses the closed-set condition $P \cap \mathbb{Z}^n = \varnothing$, such examples do not answer the question.

8 Status and remaining obstacle

The results in this note amount to partial progress only.

- Lemma 3.1 gives a correct lattice-freeness-preserving projection whenever a primitive direction z satisfies $Az \in \{-1, 0, 1\}^m$.
- The Kuhlmann–Weismantel threshold theorem supplies such directions in sufficiently high dimension as a function of a determinant bound.
- Theorem 5.1 proves a genuine dimension-free special case: if all but q variables occur through a totally unimodular matrix, then the width is at most $\mathrm{Flt}(q)$.
- Proposition 6.1 shows that the simplest determinant-preservation claim for projection is false for the Fourier–Motzkin description.

The missing ingredient for a complete positive solution of Problem 1.1 is therefore a determinant-controlled projection theory, or alternatively a structural theorem showing that every bounded-Δ_n system can be reduced to a bounded-exceptional-variable totally unimodular form. No such theorem is proved here.

References

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